

Max Pain

From bare metal to working product.

Embedded Systems Tech Lead · Principal Engineer · Software Architect

Real-time, mission- and safety-critical platforms · multi-SoC & vendor bring-up · low-latency video & rendering · C / C++ · Linux / QNX / RTOS · ISO 26262 / FuSA



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01 SUMMARY

From bare metal to working product — that is the job. The domain has changed five times in twenty-five years — virtual avionics operational with air forces, defence mission computers, autonomous-driving compute under ISO 26262 / FuSA practice, regulated medical devices, industrial control, and now AR glasses — but the requirement never has: *it is not allowed to fail*.

Across eleven roles I have owned the layers most engineers will not touch: bootloader, BSP, kernel, drivers, RTOS, multi-SoC and vendor bring-up on hardware that has never booted — then held it to hard real-time under sustained load. My working rule throughout: **solve the root cause, never patch the symptom**. No workarounds, no deadlocks, no crash dumps, every frame and every deadline met, and never a reboot as a recovery strategy.

I have hired and led two teams (~10 engineers), held principal-level titles twice, acted as technical authority across domains, and owned architecture and design end to end. Because I also build the build systems, CI/CD and release pipelines around my own platforms, I can carry a product from first power-on to a signed, shipping image without handing off the hard parts.

02 CAREER HIGHLIGHTS

- Sole real-time engineer behind the **EVA embedded virtual avionics** infrastructure — deployed worldwide and operational with leading air forces.
- Developed a **distributed multi-core defence mission computer** integrating video, graphics, analog and digital signal processing and advanced communication protocols deterministically in one system — and was the in-house authority on GHS Integrity RTOS.
- Brought up the **first central compute unit** for GM Israel's autonomous vehicle programme from scratch, and was on the core team behind its first hands-free driving demonstration.
- Took multi-sensor **video pipelines from intermittently losing frames to delivering every frame**, with latency and throughput held under sustained load.
- Currently own **full-stack bring-up of a dual-SoC AR glasses platform** — UEFI through Android userspace to signed release images, with display, cameras, sensors, power, audio, WiFi and the PCIe inter-SoC link brought up across two cores.
- Ported **Torch to QNX and Linux** and integrated CUDA into embedded compute, enabling GPU-accelerated AV machine-learning development.
- Built and led **two embedded teams (~10 engineers)** at Munters as team leader and principal software architect, defining the company's embedded and IoT platform from scratch.
- Architected a build framework at GM that cut toolchain setup **from weeks to minutes** and was adopted by many organisations across GM, spanning numerous software groups and projects.

Platform Lead — Systems, Bring-up & Release Snke XR 2025 — present

Owner of platform bring-up for a new dual-SoC AR glasses product — a **medical device** — taking it from bare metal to a signed, shippable image.

- Own every layer of the platform: UEFI bootloader display and power init, DSP and secure-world firmware, signed peripheral images, Linux kernel and per-variant device-tree overlays, Android system and vendor userspace, and the release pipeline.
- Brought up microdisplays, a multi-sensor camera array with hardware-triggered frame sync, the DSP sensor core (IMU, ambient light, magnetometer, temperature, custom virtual sensors), the PMIC power and audio paths, and per-core WiFi.
- Took the camera and display pipeline from intermittently dropping and silently losing frames to delivering every frame, with latency and throughput held under sustained load.
- Root-caused cross-core DMA misrouting and PCIe link-recovery faults; built coordinated dual-core boot and deploy over the inter-SoC link, plus self-healing wireless debug to both cores.
- Built the team's tooling: layered device-tree and overlay build system, containerised reproducible builds, a desktop DevTools application for the test fleet, an on-device C++ camera and sensor service with SDK and viewer, and 40+ executable automation procedures covering flashing, image verification, deployment and per-subsystem test sweeps.

UEFI · Linux kernel · Android/AOSP · device tree · PCIe · MIPI DSI · DSP · TrustZone · PMIC · WiFi · V4L2 · C/C++ · Electron

Principal SW Engineer — Platform & Infrastructure Baxter Healthcare 2024 — 2025

Principal-level ownership of embedded platform and release infrastructure for medical devices, inside a regulated and audited development environment.

- Redesigned, refactored and implemented fully automated CI/CD flows, adding embedded support to streamline development and deployment.
- Supported Yocto image creation for custom embedded systems on NXP platforms.
- Introduced and deployed new technology into embedded development — Azure and Azure Agents, Docker containers, Python and Bash automation.
- Led the design of a separated cloud subscription and tenancy model, giving product and organisational workloads independent, governable resource boundaries.
- Mentored teams on build tooling, compilation, linking and code organisation.

Yocto · NXP · Docker · Azure / Azure Agents · CI/CD · Python · Bash

Tech Lead — Platform & Infrastructure General Motors 2017 — 2024

Sr. RT Embedded SW Engineer (2017) → Staff Software Engineer (2019) → Tech Lead, Platform group (2019–2024). Drove platform and real-time infrastructure for advanced embedded automotive systems powering autonomous driving programmes.

- Brought up the first central compute unit and core infrastructure from scratch, enabling large-scale AV software development; established a third-party library ecosystem in Artifactory; pioneered containerisation with the programme's first on-site Docker development environment.
- Led cross-functional platform bring-ups and multi-SoC enablement on Qualcomm Ride and Nvidia Drive compute.

- Delivered Safe Linux bring-up and containerised embedded solutions aligned with FuSA / ISO 26262 safety practices.
- Evaluated and integrated DDS (RTI) and Adaptive AUTOSAR into reliable IPC middleware for high-throughput robotic applications; implemented a high-performance C++ Vehicle Data Records parser enabling ML pipelines and Elasticsearch analytics.
- Integrated CUDA into embedded compute for GPU-accelerated AI workloads and ported the Torch library to QNX and RedHat Linux, supporting early AV machine-learning work.
- Part of the core team that delivered GM Israel's first hands-free autonomous driving demonstration.
- Acted as technical authority — unblocking teams, reviewing architecture, collaborating globally with AI/ML, perception and robotics groups, and mentoring engineers on design, build systems, linking and system integration.
- Designed and implemented a modular, multi-platform end-to-end build framework reducing toolchain setup **from weeks to minutes** — supporting native Linux and cross-compilation to ARM, QNX and multiple Yocto distributions — adopted across numerous groups and projects.

C/C++ · Python · QNX · Linux (RedHat, Yocto) · Nvidia Drive · Qualcomm Ride · DDS (RTI) · AUTOSAR · CUDA · Torch · ISO 26262 · Conan · CMake · Artifactory · Jenkins · Docker / Podman

Sr. Automotive Embedded Software Solutions Engineer OSR Enterprises AG 2016 – 2017

Early-stage autonomous driving platform development — compute bring-up, sensor integration and safety standards.

- Brought up the Renesas R-Car automotive compute system on Yocto-based Automotive Grade Linux.
- Integrated Velodyne LiDAR into R-Car AGL for proof-of-concept ADAS functionality.
- Designed and implemented an MCU-based power management service controlling central compute unit power.
- Researched and evaluated automotive sensors, platforms and software stacks; built working knowledge of ADAS architecture, functional safety (ASIL A–D, ISO 26262), AUTOSAR standards and real-time OS environments.

Renesas R-Car · Yocto / AGL · QNX · GHS Integrity · PikeOS · C/C++ · Python · AUTOSAR · ISO 26262

Team Leader & Principal Software Architect — Embedded & IoT Munters 2011 – 2016

Built and led two embedded software teams (infrastructure and algorithms) while acting as principal architect for the company's core embedded and IoT software platforms.

- Hired and managed ~10 engineers across two teams, driving mentoring and project execution.
- Designed and implemented the company-wide embedded software infrastructure from scratch.
- Developed C/C++ applications, communication protocols, GUI engines and device firmware.
- Delivered advanced IoT-based control and management systems on ARM Cortex-M platforms (STM32 F0/F3/F4, NXP LPC, Atmel AVR, PIC, Intel 8051).
- Worked extensively with RTOS (FreeRTOS, CMX, ThreadX), BSP, HAL, drivers and compute board bring-ups; directed the full SDLC from requirements and architecture to production deployment.

C/C++ · ARM Cortex-M · STM32 · NXP LPC · Atmel AVR · PIC · FreeRTOS · CMX · ThreadX · BSP / HAL · IoT

Sr. RT Embedded SW Engineer Elbit Systems 2009 – 2011

Mission Computer systems for defence platforms — distributed hard-real-time software on custom multi-core hardware, where a missed deadline is a mission failure.

- Developed a complex distributed application on custom multi-core embedded platforms, integrating video, graphics, analog signal processing, digital signal processing and advanced communication protocols in one deterministic system.
- Acted as in-house expert consultant on GHS Integrity RTOS, resolving complex OS-level issues and performance bugs across multiple projects.
- Worked throughout in safety- and mission-critical environments across C/C++, Linux, GHS Integrity and VxWorks.

C/C++ · Linux · GHS Integrity · VxWorks · custom multi-core platforms · video · graphics · DSP

Sr. RT Embedded SW Engineer BVR Systems

2006 – 2009

Sole real-time embedded engineer for all bring-ups, OS integrations and infrastructure behind BVR's Embedded Virtual Avionics (EVA) training systems. Acquired by Elbit Systems in 2009.

- Performed all embedded board bring-ups (Freescale PowerPC, Intel) with GHS Integrity RTOS, ensuring stable high-performance operation.
- Designed and implemented cross-platform, POSIX-compliant infrastructure in C/C++ leveraging ACE and STL for distributed real-time simulation.
- Developed the low- and mid-level layers of Embedded Virtual Avionics, enabling avionics training systems to run on real-time embedded boards.
- Contributed to the EVA solution deployed worldwide — transforming basic and advanced trainer aircraft into high-fidelity 5th-generation virtual fighter simulators with virtual avionics, sensors and weapons, hybrid real/virtual environments simulating radar and EW, and full synchronised replay for post-flight debrief.

C/C++ · GHS Integrity · VxWorks · Windows · Freescale PowerPC · Intel · ACE · STL · POSIX

EARLY CAREER · 2001 – 2006

RT Embedded SW Engineer Albatronics

2006

New embedded board bring-up from scratch: BSP, device drivers and services on Green Hills Integrity for PowerPC boards, plus an over-the-air software-update path over JTAG. Delivered microcontroller software for MALAM and Elta defence projects and a traffic control and monitoring application for Ayalon Road Company.

C/C++ · MFC · GHS Integrity · PowerPC · JTAG

RT Embedded SW Engineer N-Trig

2005 – 2006

Embedded DSP software for the touch sensors behind early tablet PCs and flat-panel input devices — software for TI C54xx DSP processors and drivers for peripherals and input hardware, contributing to core touchscreen technology.

C · TI C54xx DSP · BIOS DSP

RT Embedded SW Engineer Mango DSP

2004 – 2005

Real-time embedded software for military ground-based defence and communication systems on PowerPC and TI C64xx DSP platforms — integrating complex communication, analog and digital signal processing modules and writing the device drivers behind them, including work for Elbit Systems.

C · TI C64xx DSP · PowerPC · VxWorks

Senior ATE Software Engineer Formtest Solutions

2001 – 2004

Automated test equipment software for military airborne and ground-based systems — ATE applications and simulators for mission-critical platforms, integrating video units, analog and digital

signal processing and complex communication systems, plus drivers for peripheral I/O devices and microcontroller programming.

C · ATEasy · ATE software

05 TECHNICAL SKILLS

embedded real-time first

REAL-TIME & MISSION-CRITICAL

Hard real-time systems · Mission-critical software
· **Safety-critical development** · Determinism & latency budgets · Concurrency, locking & lifetime
· Deadlock & starvation analysis · **FuSA** · **ISO 26262** · ASIL A-D (working knowledge) · Safe Linux bring-up · Defence & avionics mission systems · Regulated medical environments

PERFORMANCE & MEDIA

Low-latency video & streaming · Video pipelines · frame sync · Rendering · Performance optimisation
· Load & stress characterisation · Latency & throughput analysis · Analog & digital signal processing · Graphics

LANGUAGES

C · **C++** / Embedded C++ · Python · Bash / POSIX shell · Groovy

MIDDLEWARE & AI

DDS (RTI) · AUTOSAR / Adaptive AUTOSAR · IPC middleware · Torch on QNX / Linux · CUDA · ML data parsing · Elasticsearch · Claude Code · agent skills · Copilot · ChatGPT · Cursor

BUILD & RELEASE (ALSO)

CI/CD · Release engineering · Deployment automation · Reproducible builds · Toolchain management · Docker · Podman · Jenkins · Conan · CMake · Ninja · GNU Make · GCC · Artifactory · Azure · Git · GitHub · Bitbucket · repo tool · Jira

BRING-UP & LOW-LEVEL

Multi-SoC & vendor bring-up · BSP · HAL · drivers
· Device tree · UEFI / bootloader · Kernel & driver development · Yocto images · JTAG · Lauterbach · PCIe · MIPI DSI · I2C / SPI · PMIC · WiFi · TrustZone · secure boot · Hardware/software integration

OS & RTOS

Linux · Yocto · RedHat · CentOS · Ubuntu · AGL · **QNX** · Android / AOSP · **GHS Integrity** · VxWorks · PikeOS · FreeRTOS · CMX · ThreadX · uVelocity · TI BIOS DSP

SILICON

Qualcomm (Ride) · Nvidia Drive · Renesas R-Car · NXP · ARM Cortex-M · aarch64 · Freescale PowerPC · TI C54xx / C64xx DSP · STM32 · AVR · PIC · 8051

ARCHITECTURE & LEADERSHIP

Software architecture & design · **Tech lead** · **Team leader** · Team building & hiring · Architecture & code review · Mentoring · Technical authority · Full SDLC ownership · Global, cross-timezone collaboration

06 EDUCATION, SERVICE & LANGUAGES

**B.Sc. — System Analysis
& Programming**

Champlain
College · 2003

**Technician Diploma
— Radio Electronics**

Naval Technical
College — ORT ·
1996

**Military service
— IDF Navy**

Career service,
Electronics
Department ·
1999 — 2000
Electronic
Systems
Technician ·
1996 — 1999

Languages

Hebrew — native
/ bilingual
Russian — native
/ bilingual
English —
professional
working